

CHAPTER 01

INTRODUCTION

This chapter introduces the research undertaken in this thesis. It begins by presenting the background to patient queue management in public healthcare, with particular attention to the Echocardiography (ECO) unit of Sri Lankan public hospitals, before narrowing to the specific problems that motivate this study. It then states the research aim, objectives, and questions that guide the remainder of the thesis, defines the scope of the proposed system with direct reference to the three interfaces implemented in this study, discusses the significance of the research at patient, operational, managerial, national, and academic levels, and closes by outlining the structure of the chapters that follow.

1.1 Background of the Study

Efficient patient flow is one of the most consistent challenges facing public healthcare delivery worldwide, and it becomes especially critical in diagnostic and specialised units where equipment and specialist time are limited resources. Queueing theory has long been applied to model and improve service systems in healthcare, and a substantial body of research has established that structured queue management directly influences patient satisfaction, staff workload, and the overall efficiency of a hospital unit [18]. As patient volumes continue to grow in publicly funded hospitals, particularly in South Asian healthcare systems, the absence of digital tools to manage this flow has increasingly been identified as a barrier to service quality rather than a purely administrative inconvenience.

Sri Lanka's public healthcare system is widely regarded as one of the most accessible in the South Asian region, offering free treatment at the point of care through a network of national, provincial, district, and base hospitals. However, this accessibility has also produced very high patient volumes at government hospitals, and outpatient and diagnostic units frequently operate with manual, paper-based registration and queue-handling procedures. Several Sri Lankan studies have documented long and unpredictable waiting times in outpatient departments (OPDs) of public hospitals, including at the National Hospital of Sri Lanka, and have proposed operational strategies to reduce these delays [1]. Related prototypes have explored patient registration and mobile alert systems for OPDs in Sri Lankan hospitals to reduce overcrowding and improve communication with patients [2], while other local studies have investigated automated patient-flow modelling and computer-vision-based approaches to understanding congestion in OPD waiting areas [3], [4], [5]. Collectively, this body of work confirms that queue-related inefficiency in Sri Lankan public hospitals is a well-recognised and actively researched problem, but also shows that most existing local solutions address general OPD registration and communication rather than the specific operational demands of a diagnostic unit such as Echocardiography (ECO).

The Echocardiography (ECO) unit occupies a distinctive position within cardiology services because it depends simultaneously on limited numbers of trained sonographers or doctors, a restricted number of ECO machines, and examinations that vary considerably in duration depending on the clinical indication. International literature on echocardiography workflow confirms that scheduling and throughput management in echo laboratories is a recognised operational challenge, driven by growing demand for echocardiographic imaging relative to the availability of equipment and qualified staff [28], [29]. Recent international research has gone further, applying machine-learning-enhanced queueing network simulation specifically to hospital ultrasound operations, demonstrating that hybrid analytical and data-driven models can more accurately represent and improve patient flow in imaging units than traditional

first-come-first-served scheduling alone [21]. This convergence of queueing science and machine learning has become a defined research direction internationally, yet it remains largely unexplored in the context of Sri Lankan public-hospital ECO units, where registration, queue numbering, and prioritisation are still performed manually in most institutions.

At the same time, digital health has become a national policy priority in Sri Lanka. The Ministry of Health's Digital Health Blueprint sets out a roadmap for coordinated digital health investment across the public sector [8], and the National Digital Health Guidelines and Standards establish the technical and governance expectations that any new clinical-adjacent digital system is expected to meet [9]. Public-sector information systems such as the Hospital Health Information Management System (HHIMS) have already been deployed to digitise parts of hospital administration [10], and private-sector doctor-channelling platforms such as e-Channelling and Doc990 have demonstrated that Sri Lankan patients are receptive to digital appointment and queue-related services when they are made available [6], [7]. However, these national platforms are primarily oriented towards appointment booking and general hospital information management; none of them provide unit-level, real-time queue prediction or priority-based queue management tailored to a high-demand diagnostic service such as ECO.

In parallel, the global body of research applying machine learning (ML) to healthcare wait-time prediction has expanded rapidly. Studies using Random Forest, XGBoost, Gradient Boosting, and K-Nearest Neighbour models have shown that patient waiting time in outpatient and diagnostic settings can be predicted with meaningful accuracy from historical visit data, patient demographics, and temporal features [11], [12], [13], [17]. More recent work has extended this to interpretable and dissatisfaction-aware prediction models that account for the patient's perception of waiting [14], and to dynamic, self-learning queue-management systems that combine predictive analytics with real-time queue state [15], [16]. Reinforcement-learning-based approaches have also been explored for prioritising emergency and urgent cases within a shared queue, rather than relying purely on arrival order [22], [23], [24]. These developments provide the technical foundation on which the proposed system in this study is built, adapting proven ML techniques such as Random Forest [26] and XGBoost [27] to the specific operational data available in a Sri Lankan public-hospital ECO unit.

The feasibility of a patient-facing digital solution is further supported by broader trends in mobile technology adoption. Sri Lanka has one of the higher rates of mobile broadband and smartphone penetration in South Asia, and this has already enabled the growth of mobile-based healthcare services such as doctor-channelling applications and IVR-based OPD reservation lines [6], [7], [25]. This existing familiarity with mobile healthcare interactions among Sri Lankan patients reduces the adoption barrier for a dedicated queue-tracking mobile application of the kind proposed in this study, and mirrors similar mobile-augmented queue-management approaches documented in other developing-country hospital settings [25].

It is against this background — a well-documented local queueing problem, a global trend towards ML-based queue prediction, and a national digital health agenda that has not yet reached unit-level diagnostic queue management — that this study proposes the design and development of an intelligent system to manage patient flow and predict queue behaviour in the ECO section of a Sri Lankan public hospital. The system is designed around three coordinated interfaces — a patient-facing mobile application, a staff web portal, and an administrator web portal — the final implemented screens of which are presented and described in detail in Chapter 04 (Results) of this thesis.

1.2 Problem Statement

Despite the growing demand for echocardiographic examinations in Sri Lankan public hospitals, the ECO unit continues to rely on largely manual and paper-based processes for patient registration, queue handling, and reporting. Preliminary investigation conducted as part of this research, together with the wider literature on Sri Lankan OPD and hospital queue management [1], [2], [3], identified the following interrelated problems:

- Manual patient registration: Patient details are recorded on paper, which is time-consuming to search and update, prone to loss or damage, and vulnerable to duplicate registrations.
- Long and unpredictable queues: A high daily volume of patients referred from multiple hospital departments creates long queues with no reliable way to estimate waiting time, which is consistent with findings reported in Sri Lankan OPD waiting-time studies [1] and in international queueing-in-healthcare literature [18].
- No formal queue-numbering system: Patients are generally served in the order they are remembered rather than through an enforced digital ticketing mechanism, leading to disputes over queue position and repeated staff interruptions.
- Absence of appointment scheduling: Patients arrive simultaneously rather than at scheduled intervals, producing avoidable overcrowding during peak hours.
- Difficulty prioritising urgent cases: Emergency or clinically urgent patients cannot be reliably identified and moved ahead of the queue under a manual system, a gap that recent priority-queue and reinforcement-learning research has specifically sought to address in other healthcare contexts [22], [23], [24].
- Limited doctors and ECO machines: A small number of sonographers/doctors and machines create an unavoidable service bottleneck, echoing the equipment- and staff-constrained nature of echocardiography workflow reported internationally [28], [29].
- Poor communication of waiting status: Patients are not informed of expected waiting times or delays, resulting in repeated queries to staff and reduced patient satisfaction.
- Lack of real-time monitoring: Hospital management has no live visibility of queue length, station status, or daily patient volume, making it difficult to detect and respond to congestion as it happens.
- Inaccurate resource and staff planning: Without historical trend analysis, staffing and resource-allocation decisions are made reactively rather than being informed by predictable demand patterns, despite strong evidence from international demand-forecasting research that patient volumes can be predicted from historical data with useful accuracy [30].
- Paper-based reporting and absence of historical analysis: Monthly and annual performance reports are compiled manually, which is time-consuming, error-prone, and does not support trend or pattern analysis for future planning.

These problems are not unique to Sri Lanka; however, the specific combination of a paper-based public-hospital environment, a specialised and resource-constrained diagnostic unit, and the absence of any

locally deployed predictive queue-management tool represents a distinct and, to date, insufficiently addressed research gap. Existing Sri Lankan prototypes have targeted general OPD registration and SMS-based alerts [2], and existing national platforms such as e-Channelling and HHIMS operate at the level of appointment booking and hospital-wide information management rather than real-time, ML-driven queue prediction within a single diagnostic unit [6], [9]. Likewise, while international research has begun to combine machine learning with queueing models for imaging services [21], this has not yet been examined within a low-resource, public-hospital ECO setting such as that found in Sri Lanka. This study is therefore motivated by the need for an integrated system that can digitally manage patient registration, automatically generate and track queue numbers, prioritise urgent cases, predict waiting times and daily patient demand using machine learning, and provide real-time analytical dashboards for hospital staff and management — specifically for the ECO unit of a Sri Lankan public hospital.

The practical cost of leaving this problem unaddressed extends beyond patient inconvenience. Unpredictable queues in a diagnostic unit such as ECO can lead to missed or delayed examinations, extended patient stays elsewhere in the hospital while patients wait for an ECO slot, avoidable overtime or congestion for clinical staff, and management decisions on staffing and equipment procurement that are based on impression rather than evidence, consistent with the reactive resource-planning weaknesses reported in the wider queueing-in-healthcare literature [18], [30]. Addressing these problems through an integrated, ML-supported digital system therefore has the potential to produce measurable operational improvements — reduced waiting time, improved patient satisfaction, more efficient staff utilisation, and evidence-based resource planning — in a setting where the existing literature confirms these gains have already been observed following digital or predictive queue interventions elsewhere [11], [15], [22].

1.3 Research Aim and Objectives

1.3.1 Research Aim

The aim of this study is to design, develop, and evaluate an intelligent, machine-learning-supported system that digitally manages patient registration and queueing, prioritises urgent cases, predicts patient waiting time and queue congestion, and provides real-time operational visibility for staff and administrators in the Echocardiography (ECO) unit of a Sri Lankan public hospital.

1.3.2 Research Objectives

1. To critically review existing local and international literature on hospital queue management, queueing theory, and machine-learning-based wait-time and demand prediction, in order to identify the specific research gap relevant to Sri Lankan public-hospital ECO units [1]–[24].
2. To design and implement a digital patient registration and automated queue-ticket generation module, corresponding to the patient-facing Home Dashboard, Join Queue, and Live Ticket screens implemented in this study.
3. To design and implement a priority-based queue management module that allows staff to identify, flag, and advance urgent cases ahead of routine patients, corresponding to the priority-tagging functionality on the Staff Queue Management screen, informed by priority-queue and reinforcement-learning approaches reported in the literature [22], [23], [24].

4. To develop a machine-learning model capable of predicting patient waiting time and hourly/daily queue volume for the ECO unit using historical and real-time queue data, drawing on established algorithms such as Random Forest and XGBoost [26], [27] and informed by comparable healthcare wait-time prediction studies [11]–[17], corresponding to the Queue Prediction screen on the staff portal.
5. To implement real-time queue tracking and status communication for patients, so that they can view their live position, estimated waiting time, and stage of their visit, corresponding to the Live Queue Tracking screen.
6. To design and implement administrator-level oversight tools, including station/room management, staff-account management, an audit log, system settings, and analytical reporting dashboards, corresponding to the Administrator Web Portal screens implemented in this study.
7. To evaluate the functionality, usability, and predictive accuracy of the developed system, and to document the results and their implications for public-hospital ECO queue management in Sri Lanka.

1.4 Research Questions

Based on the problem statement and research objectives above, this study seeks to answer the following research questions:

8. What are the primary causes of long and unpredictable patient waiting times in the ECO unit of a Sri Lankan public hospital, and how do they compare with causes documented in local and international literature?
9. To what extent can a digital registration and automated ticketing system reduce manual administrative overhead and queue-position disputes in a public-hospital ECO unit?
10. How effectively can a priority-based queue-management mechanism identify and advance clinically urgent patients without significantly disadvantaging routine patients in the same queue?
11. How accurately can machine-learning models predict patient waiting time and hourly/daily queue volume for the ECO unit using available historical and real-time data?
12. What real-time information and analytical tools do hospital staff and administrators require to effectively monitor and manage ECO-unit queue congestion, staffing, and resource allocation?
13. How do patients and hospital staff perceive the usability and usefulness of the proposed mobile, staff, and administrator interfaces in a real or simulated ECO-unit setting?

1.5 Scope of the Study

This study is scoped to the design, development, and evaluation of a queue-prediction and patient-flow-management system for a single diagnostic unit — Echocardiography (ECO) — within the context of a Sri Lankan general/public hospital. The system is implemented as three coordinated interfaces, each supporting a distinct user role, and the scope of this study is defined precisely by the final implemented screens of those interfaces, presented in full in Chapter 04 (Results) of this thesis.

The Patient (User) Mobile Application is in scope and provides five core screens: a Home Dashboard summarising the patient's current ticket and quick actions; a Join Queue screen for service/department selection; a Live Ticket screen confirming the assigned queue number and status; a Live Queue Tracking screen showing a staged, colour-coded timeline of the visit; and a Visit History screen listing past visits with status badges.

The Staff Web Portal is in scope and provides seven core screens: a Today's Overview dashboard with key-performance-indicator cards and a patient-volume chart; a Queue Management screen for calling, serving, skipping, and prioritising patients; a Queue Prediction screen presenting machine-learning-generated forecasts of patient influx, average wait time, and peak hours; a Patient Search screen; a Patient Detail screen with visit history and a messaging panel; a Reports screen with exportable operational statistics; and a Register New Patient modal for adding walk-in patients directly into the queue.

The Administrator Web Portal is in scope and provides eight core screens: a Today's Unit Overview dashboard combining unit-wide key-performance-indicator cards, station status, and system alerts; a Room/Station Management grid; a Staff Accounts management table; an Add-Staff-Member modal; an Audit Log for recording system and user actions; a Settings screen for role-based permissions and security configuration; a Register Station modal; and an Add Echo Room modal.

The machine-learning component of the system is limited to predicting patient waiting time and queue volume/congestion for the ECO unit based on historical and real-time operational data (such as arrival patterns, time of day, day of week, and station capacity); it does not extend to clinical diagnosis, image analysis, or interpretation of echocardiographic results, which remain outside the scope of this study. Similarly, the system is scoped to queue and patient-flow management within the ECO unit specifically, and does not attempt to replace the hospital's existing Hospital Health Information Management System (HHIMS) [10] or national e-Channelling appointment infrastructure [6]; rather, it is intended to complement these systems at the unit level, in a gap that they do not currently address. Billing, insurance processing, and full electronic health-record functionality are likewise excluded from the scope of this study. The evaluation of the system is limited to functional testing, usability assessment, and predictive-accuracy testing using available and/or simulated ECO-unit queue data, rather than a full multi-hospital clinical trial.

1.6 Significance of the Study

This study is significant at the patient, operational, managerial, and academic levels.

For patients, the proposed system replaces uncertainty with visibility: rather than waiting without information, patients using the mobile application can view their live ticket number, an estimated waiting time, and a staged tracking timeline of their visit, directly addressing the poor communication of waiting status identified in the problem statement and in Sri Lankan OPD literature [1], [2].

For front-desk and clinical staff, the Staff Web Portal is significant because it consolidates registration, live queue management, priority handling, patient search, and reporting into a single digital workflow, reducing the administrative burden associated with manual, paper-based processes and supporting the kind of priority handling for urgent cases that manual systems in Sri Lankan public hospitals currently struggle to guarantee [22], [23], [24].

For hospital administrators and management, the Administrator Web Portal provides, for the first time at unit level, real-time oversight of station status, staff accounts, system alerts, and audit activity, together with reporting and insight tools that support data-driven resource and staffing decisions rather than the reactive planning described in the problem statement. This directly responds to the international evidence that historical and real-time data can meaningfully inform hospital resource and demand planning [30].

At a national level, this study contributes to Sri Lanka's digital health agenda by extending digital-health thinking from hospital-wide information management and appointment booking — the current focus of platforms such as HHIMS and e-Channelling [6], [9], [10] — down to real-time, predictive, unit-level queue management, an area explicitly identified as under-addressed in the existing literature and national digital health guidance [8].

Academically, this study contributes to the growing international literature that combines machine learning with queueing and patient-flow management in healthcare [11]–[17], [21]–[24], [26], [27], by applying and evaluating these techniques within a previously unexamined context: a resource-constrained, diagnostic (echocardiography) unit within a Sri Lankan public hospital. In doing so, it provides an empirical case study that can inform both future Sri Lankan digital-health research and broader research on ML-based queue management in low-resource healthcare settings internationally.

More broadly, the significance of this study extends to other resource-constrained diagnostic units beyond ECO — such as radiology, laboratory, and endoscopy services — that share the same underlying combination of limited equipment, limited specialist time, and manual queue handling. Because the system architecture proposed in this study (digital registration, priority queueing, ML-based prediction, real-time tracking, and administrative oversight) is not clinically specific to echocardiography, the findings of this research are intended to offer a transferable model that other diagnostic units within Sri Lankan public hospitals, and similarly resourced public-health systems in other developing countries, could adapt for their own queue-management needs.

1.7 Structure of the Thesis

The remainder of this thesis is organised into four further chapters, structured as follows:

Chapter 02 – Literature Review: reviews Sri Lankan and international literature on hospital queue management, queueing theory, existing digital health and channelling platforms, and machine-learning approaches to wait-time and demand prediction, and concludes by consolidating the specific research gap this study addresses.

Chapter 03 – Methodology: describes the research design, data-collection approach, the machine-learning techniques and tools used to build the prediction model, the system-development approach for the three interfaces, and the ethical considerations observed during data collection and testing.

Chapter 04 – Results: presents the final implemented system, including the complete set of Patient Mobile Application, Staff Web Portal, and Administrator Web Portal screens, together with the results of the machine-learning prediction model.

Chapter 05 – Discussion: interprets the results in relation to the research objectives and questions set out in this chapter, discusses the study's limitations, and concludes with recommendations for future work.

Together, these chapters build directly on the background, problem statement, and objectives established in this introductory chapter, and culminate in the fully implemented, three-role queue-management and prediction system for the ECO unit that is documented in Chapter 04.

1.8 Chapter Summary

This chapter has established the foundation for the remainder of the thesis. It has shown that patient queueing in the ECO unit of Sri Lankan public hospitals is a well-documented but still largely unaddressed problem, situated between an active body of local OPD-focused research [1]–[5], a maturing international literature on machine-learning-based queue and wait-time prediction [11]–[17], [21]–[24], [26], [27], and a national digital health agenda that has not yet reached real-time, unit-level queue management [6], [8]–[10]. From this background, the chapter derived a detailed problem statement, a clear research aim and seven supporting objectives, six research questions, a scope defined precisely by the twenty implemented screens across the Patient Mobile Application, Staff Web Portal, and Administrator Web Portal, and the significance of the study at patient, staff, management, national, and academic levels. The next chapter, Chapter 02, builds on this foundation with a detailed and critical review of the literature introduced here.

CHAPTER 02

LITERATURE REVIEW

2.1 Introduction

This chapter presents a critical review of the literature relevant to the design and development of an intelligent patient-flow and queue-management system for Echocardiography (ECO) units in Sri Lankan public hospitals. The review draws on thirty sources spanning four broad bodies of knowledge: (i) local, Sri Lanka-specific studies and national digital-health policy documents that describe the current state of outpatient department (OPD) operations and digitisation efforts; (ii) international literature on machine learning (ML) approaches to predicting patient wait times and queue behaviour in clinical settings; (iii) foundational queueing-theory and simulation research that provides the mathematical and methodological basis for modelling patient flow; and (iv) domain-specific literature on echocardiography workflow optimisation, priority queueing, Internet-of-Things (IoT)/mobile queue systems, and hospital demand forecasting.

The chapter is organised as follows. Section 2.2 reviews the theoretical foundations of queueing theory and simulation as applied to healthcare. Section 2.3 examines the Sri Lankan context, including national digital-health policy and empirical OPD studies. Section 2.4 reviews global machine-learning approaches to wait-time and queue-length prediction, together with the core algorithmic foundations (Random Forest and XGBoost) that inform the predictive component of the proposed system. Section 2.5 discusses priority queueing and AI-driven hospital queue-management systems. Section 2.6 reviews IoT and mobile-based smart-queue architectures. Section 2.7 focuses specifically on echocardiography workflow optimisation. Section 2.8 reviews hospital resource-demand forecasting literature. Section 2.9 provides a critical, cross-cutting synthesis of the reviewed literature, and Section 2.10 consolidates the research gaps that motivate the present study, mapping each gap to the corresponding feature of the proposed system. Section 2.11 summarises the chapter.

2.2 Theoretical Foundations: Queueing Theory and Simulation in Healthcare

Queueing theory provides the mathematical foundation on which almost all quantitative queue-management research is built, and it is therefore the natural starting point for this review. Fomundam and Herrmann [18] provide a widely cited survey of queueing-theory applications in healthcare, classifying prior work into categories such as bed-capacity planning, personnel and staff scheduling, and outpatient waiting-time analysis. Their survey establishes the standard vocabulary of arrival and service distributions (e.g., M/M/1, M/M/c and M/G/1 systems) that later, more applied studies build upon, and it highlights an early and persistent observation in the field: that hospital arrival and service processes rarely satisfy the strict statistical assumptions of classical queueing models, which motivates the shift toward simulation and, more recently, machine-learning-based approaches.

Building on these theoretical foundations, a study on queue management in a Malaysian public hospital [19] demonstrates how classical single- and multi-server queueing models can be adapted to a resource-constrained, developing-country public hospital setting that is structurally comparable to Sri Lanka. The study shows that even relatively simple queueing interventions, when combined with basic digitisation of patient registration, can meaningfully reduce perceived and actual waiting times without requiring additional clinical staff or equipment. This finding is particularly relevant to the resource constraints described for ECO units in Sri Lankan public hospitals, where the number of doctors and echocardiography machines is fixed in the short term.

A more recent study applies queueing theory and simulation specifically to emergency-department waiting times through an Accumulative Priority Queue with a Finite Horizon [20]. This model addresses a key weakness of both pure first-come-first-served (FCFS) queues and static priority queues: patients with lower priority can be indefinitely delayed if higher-priority patients continue to arrive. The finite-horizon accumulative priority mechanism allows a patient's effective priority to increase the longer they wait, guaranteeing an upper bound on delay while still allowing urgent cases to be seen sooner. This is directly relevant to the ECO context, where urgent cardiac referrals must be prioritised over routine outpatient echocardiograms without indefinitely starving the routine queue.

The study most methodologically similar to the present research is a high-fidelity, machine-learning-enhanced queueing-network simulation of hospital ultrasound operations [21]. Ultrasound, like echocardiography, is a multi-stage, equipment- and operator-constrained imaging service, and the study combines discrete-event simulation of the queueing network with ML-based surrogate models of patient arrival and service-time distributions. This hybrid simulation-plus-ML approach demonstrates that predictive models can be embedded inside a queueing-network simulation to produce far more realistic estimates of waiting time and throughput than either technique alone, and it directly informs the architectural approach adopted in the present study, which likewise couples an ML-based wait-time predictor with a queue-management engine rather than treating prediction and queue control as separate concerns.

Table 2.1: Summary of queueing-theory and simulation foundations reviewed

Ref.	Study focus	Method	Relevance to this study
[18]	Survey of queueing theory in healthcare	Literature survey / taxonomy	Provides the standard queueing vocabulary and model taxonomy used

Ref.	Study focus	Method	Relevance to this study
			throughout this thesis
[19]	Public-hospital queue management, Malaysia	Applied queueing model + case study	Confirms feasibility of simple queueing interventions in resource-constrained public hospitals
[20]	Emergency-department waiting time reduction	Accumulative priority queue, finite horizon	Basis for the bounded-delay priority discipline proposed for urgent ECO cases
[21]	Hospital ultrasound operations	ML-enhanced queueing-network simulation	Closest architectural precedent for combining ML prediction with queue simulation

2.3 Digital Health Infrastructure and OPD Queue Management in Sri Lanka

At the national-policy level, the Sri Lanka Digital Health Blueprint [8], published by the Ministry of Health, Nutrition and Indigenous Medicine, sets out the government's strategic vision for digitising health services, including outpatient and diagnostic services, and identifies interoperability and data standardisation as key enablers. This is reinforced at a technical level by the National Digital Health Guidelines and Standards, version 2.0 [9], which specifies the interoperability, data-exchange and system-integration requirements that any new digital health system deployed in a Sri Lankan public hospital, including the proposed ECO queue-management system, would be expected to satisfy. A retrospective review of Sri Lanka's Health Information Management System (HHIMS) covering the period 1998-2017 [10] documents the country's two-decade digitisation journey, and highlights recurring adoption barriers such as inconsistent infrastructure across hospitals, limited staff digital literacy, and fragmented, department-level rather than hospital-wide system deployment. These barriers are directly relevant to the design and rollout strategy of the proposed system.

At the empirical level, several Sri Lankan studies have examined OPD waiting-time problems directly. Dilrukshi et al. [1] propose a strategy to reduce waiting time at the OPD of the National Hospital of Sri Lanka, using simulation-based bottleneck analysis of the registration and consultation process to recommend targeted resource reallocation. While effective as a diagnostic tool, the study does not propose an ongoing digital system; it produces a one-off set of recommendations rather than a deployable, continuously operating queue-management platform. Akmal Jahan et al. [2] describe OPD-PMMAS, a prototype patient-management and mobile-alert system developed for OPDs in Sri Lankan hospitals, which digitises patient registration and issues SMS-based alerts to inform patients of their queue status. OPD-PMMAS is conceptually close to the notification and mobile-app components of the system proposed in this thesis, but it remains at prototype scale, is not ECO-specific, and does not

incorporate any predictive or machine-learning capability for estimating waiting time or forecasting demand.

A separate strand of Sri Lankan research has modelled OPD patient flow using discrete-event simulation techniques [3], quantifying how changes in staffing levels and service-station allocation affect aggregate waiting time; however, as with [1], the output is a simulation model for offline analysis rather than a live operational system. Two further Sri Lankan studies apply computer-vision techniques to OPD operations: one evaluates human-detection algorithms for automatically counting patients in OPD waiting areas [4], and a companion study applies nameboard detection and text recognition to automatically identify clinic and department signage [5]. Both studies demonstrate that computer-vision techniques can automate data capture that would otherwise require manual staff effort, which is relevant to the real-time queue-length monitoring and way-finding functions envisaged for the proposed system, although neither study integrates its output into an end-to-end queue-management or prediction pipeline.

Finally, two studies describe existing, deployed digital-health services in Sri Lanka that represent the current state of practice against which the proposed system can be differentiated. The eChannelling digital OPD reservation service [6], launched by eChannelling PLC in partnership with ICTA Sri Lanka, allows patients to book OPD appointments at district hospitals online, but it functions purely as a booking/scheduling channel and does not perform live queue tracking, priority management, or predictive waiting-time estimation within the hospital on the day of the visit. Similarly, the Doc990 mobile service documented by GSMA Mobile for Development [7] uses basic mobile channels (voice/SMS) to improve patient access to health information and services in Sri Lanka, illustrating the importance of low-friction mobile access in a context where not all patients own smartphones, but again without any queue-prediction or in-hospital flow-management functionality. Together, these studies confirm that while digital OPD services exist in Sri Lanka, none combines patient registration, real-time priority-aware queue management, and ML-based wait-time prediction into a single system targeted at a specific diagnostic unit such as echocardiography.

Table 2.2: Sri Lanka-specific studies and identified limitations relative to the proposed system

Ref.	Study / system	Scope	Limitation relative to the proposed ECO system
[1]	National Hospital OPD waiting-time strategy	Simulation-based diagnostic study	One-off analysis, not a deployable system; not ECO-specific
[2]	OPD-PMMAS prototype	Patient management + SMS alerts	Prototype scale; no ML prediction; not ECO-specific
[3]	OPD patient-flow simulation	Discrete-event simulation	Offline model only; no live/operational component
[4]	Human-detection for OPD	Computer-vision patient counting	Not integrated with queue prediction or

Ref.	Study / system	Scope	Limitation relative to the proposed ECO system
			management
[5]	Nameboard detection / OCR	Computer-vision way-finding	Not integrated with queue prediction or management
[6]	eChannelling digital OPD reservation	Deployed appointment-booking service	Booking only; no in-hospital live queue or prediction
[7]	Doc990 mobile health service	SMS/voice patient-access channel	General health access; no queue management

2.4 Machine Learning Approaches to Wait-Time and Queue-Length Prediction

A substantial and growing international literature applies supervised machine-learning techniques to predict outpatient and clinic waiting times, and this literature forms the primary methodological basis for the predictive component of the proposed system. Li et al. [11] predict outpatient waiting time in a tertiary children's hospital using a range of ML models trained on features such as registration time, department, and physician workload, and report that tree-based ensemble methods consistently outperform traditional linear regression approaches. Their results validate the feasibility of ML-based wait-time prediction in large, complex hospital settings with multiple departments and physicians, a structural profile that is comparable to the multi-doctor, multi-machine environment of an ECO unit.

A related study published in *Informatics in Medicine Unlocked* [12] compares several machine-learning algorithms for predicting wait times in an outpatient clinic and emphasises the importance of feature engineering, particularly features that capture a patient's real-time position in the queue and the time of day, over simply adding more algorithmic complexity. A 2024 preprint [13] extends this line of work to predicting patient wait times together with appointment delays, which is relevant because unpredictable delays and no-shows directly affect the accuracy of any real-time queue-length estimate; a robust ECO queue-prediction model must therefore account for both service-time variability and non-attendance.

An important methodological contribution comes from a 2024 study that frames wait-time prediction as a dissatisfaction-considered, interpretable machine-learning problem [14]. Rather than optimising only for predictive accuracy, the study explicitly incorporates patient-reported dissatisfaction as an outcome and stresses model interpretability (for example, through feature-importance or SHAP-based explanations) so that clinical and administrative stakeholders can trust and act on the model's predictions. This directly informs the design choice, adopted in the present study, to favour interpretable tree-ensemble models over opaque deep-learning architectures for the ECO wait-time predictor.

Guo et al. [15] propose machine-learning-based prediction models embedded within a dynamic hospital queue-management system, rather than as a standalone offline prediction tool, which is conceptually the closest precedent to the integrated prediction-plus-management architecture proposed in this thesis. A separate JMIR Medical Informatics study conducted in a paediatric hospital [16] validates ML-based wait-

time prediction on a large-scale, real-world retrospective dataset, providing evidence that such models can generalise beyond small pilot studies to practical, real-world deployment. Finally, a 2025 comparative study published in the International Journal of Engineering Trends and Technology [17] benchmarks multiple machine-learning algorithms against one another for predicting consultation wait times in outpatient clinics, providing a direct evidence base for algorithm selection.

The consistent finding across [11], [15], [16] and [17] is that ensemble tree-based methods, and gradient-boosted trees in particular, tend to outperform both simple regression baselines and, in several cases, more complex neural architectures on structured, tabular hospital operations data of the kind that an ECO unit would generate (patient demographics, referral source, time of arrival, doctor/machine allocation, and historical service durations). This motivates a closer examination of the two core algorithms that underlie most of these studies and that are adopted as the primary candidate algorithms for the predictive module of the proposed system.

2.4.1 Core Machine Learning Algorithms: Random Forest and XGBoost

Breiman's Random Forest algorithm [26] is a bagging-based ensemble method that constructs a large number of de-correlated decision trees over bootstrap-resampled subsets of the training data and features, and aggregates their predictions by averaging (for regression) or majority voting (for classification). Random Forests are well suited to noisy, mixed-type healthcare operational data because they are relatively insensitive to outliers and missing values, require minimal feature scaling or preprocessing, and provide a natural measure of feature importance that supports the interpretability requirement identified by [14]. These properties make Random Forest a strong baseline candidate for predicting ECO queue length and waiting time from heterogeneous inputs such as patient category, referral urgency, time of day, and current queue occupancy.

Chen and Guestrin's XGBoost [27] is a scalable, regularised implementation of gradient-boosted decision trees that builds an ensemble of trees sequentially, with each new tree trained to correct the residual errors of the preceding ensemble. XGBoost incorporates second-order gradient information, L1/L2 regularisation, and efficient handling of sparse data, and has become one of the dominant algorithms for structured/tabular prediction tasks across many domains, including several of the wait-time prediction studies reviewed above (notably [11] and [17], both of which report gradient-boosted trees among their top-performing models). Given its consistent superiority on structured healthcare datasets in the literature reviewed, and its practical advantages in training speed and predictive accuracy over plain Random Forests when properly tuned, XGBoost is adopted as the primary candidate algorithm for the ECO queue-length and wait-time prediction module in the proposed system, with Random Forest retained as an interpretable benchmark model.

Table 2.3: Comparative summary of machine-learning wait-time prediction studies

Ref.	Clinical context	Algorithm(s) evaluated	Reported best performer	Relevance to ECO prediction module
[11]	Paediatric tertiary hospital, multi-department	Linear regression, tree ensembles	T r e e - b a s e d ensembles	Validates ML feasibility in complex, multi-doctor settings

Ref.	Clinical context	Algorithm(s) evaluated	Reported best performer	Relevance to ECO prediction module
[12]	General outpatient clinic	Multiple ML algorithms	Not fully specified; emphasises feature engineering	Informs feature-set design (queue position, time of day)
[13]	Outpatient clinics	ML regression/classification	Not fully specified	Highlights need to model delays / no-shows
[14]	Outpatient department	Interpretable ML (e.g. tree-based, SHAP)	Interpretable ensemble model	Basis for interpretability requirement
[15]	Hospital-wide dynamic queue	ML prediction embedded in queue system	ML-based dynamic models	Closest precedent for integrated architecture
[16]	Paediatric hospital, real-world	ML models, retrospective validation	Not fully specified	Evidence of real-world deployability
[17]	Outpatient consultation	Comparative ML benchmark	Gradient-boosted trees	Direct evidence for XGBoost selection
[26]	General (methodological)	Random Forest	-	Interpretable benchmark algorithm
[27]	General (methodological)	XGBoost	-	Primary candidate prediction algorithm

2.5 Priority Queueing and AI-Driven Hospital Queue Management

Beyond prediction alone, several recent studies combine artificial intelligence with priority-queueing logic to actively manage, rather than merely forecast, hospital queues. A 2024 study optimises hospital queue management using a priority-queue algorithm combined with reinforcement learning for emergency-service prioritisation [22], showing that a reinforcement-learning agent can learn dynamic prioritisation policies that adapt to real-time queue conditions, outperforming static priority-level rules of the kind used in most existing systems, including the finite-horizon accumulative priority model of [20]. This suggests that a future extension of the proposed ECO system could move from static priority weighting (e.g., urgent versus routine referral) toward a learned, adaptive prioritisation policy.

MediQueue [23], a 2025 ML-driven hospital queue-management system, is architecturally close to the system proposed in this thesis: it combines predictive modelling with active queue orchestration in a

single deployable system rather than treating prediction as a separate analytical exercise. As such, MediQueue serves as a useful comparative benchmark for the system-design decisions and evaluation metrics (e.g., prediction accuracy, average wait-time reduction, queue-length error) adopted later in this thesis. Similarly, Kavitha and Maragathasundari [24] propose AI-enabled queueing-theory applications for hospital patient-flow optimisation, integrating classical queueing performance metrics (e.g., utilisation, average queue length, average waiting time) with AI-based optimisation techniques. Their work reinforces a recurring theme across this section of the literature: that combining classical queueing theory with modern AI/ML techniques consistently outperforms either approach used in isolation, which is the design philosophy adopted for the ECO queue-management engine proposed in this study.

2.6 IoT and Mobile-Based Smart Queue Systems

The patient-facing component of the proposed system, comprising a mobile application through which patients join a queue, receive a live ticket number, and track their position in real time, draws on the broader literature on IoT and mobile-based smart-queue systems. An early but influential study describes a smart mobile system for the real-time tracking and management of service queues [25]. Although developed for general service industries rather than healthcare specifically, the system's core architectural pattern, comprising remote ticket issuance, push-notification-based status updates, and live queue-position tracking on a patient's own device, is directly transferable to the healthcare context and closely mirrors the Home Dashboard, Join Queue, Live Ticket, and Live Queue Tracking screens implemented for the patient mobile application in the present study. The absence of a healthcare-specific, ML-integrated equivalent of this architecture in the literature reviewed represents a further gap that the proposed system addresses.

2.7 Workflow Optimisation in Echocardiography Services

Literature focused specifically on echocardiography workflow, as distinct from general OPD or emergency-department queue management, is comparatively scarce, which itself constitutes an important motivation for this study. A review published in *US Cardiology Review* [28] identifies the principal bottlenecks that constrain throughput in echocardiography laboratories, including scheduling inefficiencies, sonographer and physician availability, equipment turnaround time between studies, and reporting/sign-off delays. The review is largely descriptive and does not propose or evaluate a specific digital or predictive intervention.

Solis et al. [29] present a more applied study on optimising echocardiography workflow to increase patient access to high-quality care, proposing process-level changes such as protocol standardisation, appointment triage, and more efficient scheduling of studies by complexity, in order to increase throughput without requiring additional machines or staff. This is the most clinically and operationally relevant precedent for the ECO-specific focus of the present research. However, neither [28] nor [29] incorporates machine-learning-based prediction of waiting time or queue length, nor a real-time digital queue-management system with a patient-facing mobile interface; both rely on manual process redesign rather than an automated, data-driven system. This absence of a predictive, technology-enabled solution specifically for echocardiography services represents one of the clearest gaps addressed by this thesis.

2.8 Demand Forecasting for Hospital Resource Planning

Finally, a 2025 study published in the Gulf Journal of Advance Business Research examines the forecasting of hospital resource demand using a combination of time-series and machine-learning models [30]. Although the study addresses hospital-wide resource demand (such as bed occupancy and staffing needs) rather than a single diagnostic unit, its methodological approach, combining classical time-series techniques with machine-learning models to forecast future demand, is directly transferable to forecasting daily and weekly patient volumes for an ECO unit. This informs the design of the Queue Length Forecasting feature and the administrator-facing analytics dashboard proposed in this thesis, extending general hospital demand-forecasting principles to the specific, previously unaddressed context of echocardiography services in a Sri Lankan public hospital.

2.9 Critical Analysis and Synthesis

Read together, the thirty sources reviewed in this chapter reveal four consistent patterns. First, the Sri Lanka-specific literature ([1]-[7]) demonstrates clear awareness of OPD queue-management problems and has produced valuable prototypes and analytical studies, but almost none of these systems incorporate predictive machine learning, and none is specific to echocardiography or any single diagnostic-imaging unit; most also remain at prototype or one-off analytical scale rather than being deployed as production systems, a gap partially explained by the infrastructure and adoption barriers documented in the national HHIMS review [10].

Second, the international machine-learning literature ([11]-[17], [26], [27]) provides strong, repeated empirical evidence that tree-based ensemble methods, particularly gradient-boosted trees, can predict outpatient wait times with useful accuracy, and that interpretability matters for clinical adoption [14]. However, this literature is overwhelmingly generic to outpatient clinics in general or to specific non-cardiac specialties (e.g., paediatrics [11], [16]), and none of the reviewed studies addresses echocardiography or any comparably equipment-constrained, multi-stage imaging service.

Third, the queueing-theory, simulation, and priority-queueing literature ([18]-[24]) provides strong theoretical and methodological tools, including finite-horizon accumulative priority disciplines [20] and reinforcement-learning-based dynamic prioritisation [22], for managing urgent and routine cases fairly and efficiently. The ML-enhanced ultrasound simulation study [21] and the MediQueue system [23] demonstrate that combining these theoretical tools with ML prediction in a single deployable system is both feasible and effective, but, again, no reviewed study applies this combined approach to echocardiography specifically, nor within the Sri Lankan public-hospital context.

Fourth, the domain-specific echocardiography literature ([28], [29]) confirms that ECO-specific workflow bottlenecks are real and clinically significant, but the proposed remedies are limited to manual process redesign rather than automated, predictive, technology-enabled solutions. The IoT/mobile smart-queue literature [25] and hospital demand-forecasting literature [30] each provide transferable architectural and analytical components (real-time mobile tracking; time-series/ML demand forecasting) that have not previously been assembled together, nor applied to an ECO unit.

2.10 Identification of Research Gaps

Synthesising the critical analysis in Section 2.9 with the operational problems identified in the research proposal (manual paper-based registration, unpredictable and long queues, absence of a queue-numbering system, lack of appointment scheduling, difficulty prioritising urgent cases, absence of real-

time monitoring, and inaccurate, reactive resource planning), five specific research gaps can be identified in the existing literature, none of which is fully addressed by any single reviewed study. Table 2.4 maps each identified gap to the supporting literature that reveals it, and to the corresponding feature of the system proposed in this thesis.

Table 2.4: Research gaps identified from the literature and corresponding proposed-system features

#	Identified research gap	Supporting literature	Corresponding proposed-system feature
G1	No Sri Lankan OPD/ECO system combines digital registration with ML-based wait-time and queue-length prediction	[1]-[7], [11]-[17]	Digital Patient Registration + Waiting Time Prediction (ML)
G2	Existing priority-queueing and reinforcement-learning approaches are not applied within an ECO-specific, resource-constrained public-hospital setting	[20], [22], [23], [24]	Priority Queue Management for Urgent Cardiac Cases
G3	Echocardiography workflow-optimisation studies rely on manual process redesign rather than automated, real-time, data-driven queue management	[28], [29]	Real-Time Queue Tracking + Patient Flow Analysis Dashboard
G4	No reviewed system combines a patient-facing mobile smart-queue interface with an ML-enhanced queueing-network back end for a specific diagnostic-imaging service	[21], [25]	Patient Mobile Application (Join Queue, Live Ticket, Live Tracking)
G5	Hospital resource-demand forecasting has not been applied to daily/weekly ECO patient-volume prediction to support proactive staffing	[30]	Queue Length Forecasting (ML) + Staff and Resource Monitoring

#	Identified research gap	Supporting literature	Corresponding proposed-system feature
	and scheduling		

2.11 Chapter Summary

This chapter has reviewed thirty sources spanning Sri Lanka-specific OPD and digital-health studies, global machine-learning approaches to wait-time and queue-length prediction, foundational queueing-theory and simulation research, priority-queueing and AI-driven hospital queue-management systems, IoT/mobile smart-queue architectures, echocardiography-specific workflow-optimisation studies, and hospital resource-demand forecasting. The critical synthesis in Section 2.9 shows that, while each individual component required for the proposed system (digital registration, ML-based prediction, priority queueing, real-time mobile tracking, and demand forecasting) has precedent somewhere in the literature, no existing study or system integrates all of these components together, and none does so specifically for echocardiography services within a Sri Lankan public-hospital context. Section 2.10 formalised this observation into five specific, literature-grounded research gaps and mapped each to a corresponding feature of the system proposed in this thesis. These gaps directly motivate the research aim, objectives, and system design described in Chapter 1 and operationalised through the methodology presented in Chapter 3.

CHAPTER 03

METHODOLOGY

3.1 Introduction

This chapter presents the research methodology adopted for the design, development, and evaluation of an intelligent patient-flow and queue-management system for the Echocardiography (ECO) unit of a Sri Lankan public hospital. As established in Chapter 2, the study addresses a set of literature-grounded research gaps by combining digital patient registration, machine-learning-based wait-time and queue-length prediction, priority-aware queue management, and real-time mobile/web interfaces into a single, integrated system, delivered through the Patient Mobile Application, the Staff Web Portal, and the Administrator Web Portal.

This chapter is submitted as the methodology component of Interim Submission 01 and therefore focuses on the research design, the data-collection strategy, the development tools and technologies, and the ethical considerations that govern the study, consistent with the chapter breakdown specified in the project guidelines. The detailed presentation of the resulting system implementation (the finalised patient, staff, and administrator interface screens) is reported separately in Chapter 4 (Results), and the evaluation of those results is discussed in Chapter 5 (Discussion); this chapter therefore describes how the system was planned, designed, and built, and how the supporting data and evaluation instruments were collected, rather than the outcomes themselves.

The remainder of the chapter is organised as follows: Section 3.2 describes the overall research design; Section 3.3 describes the system-development methodology adopted; Section 3.4 presents the functional and non-functional requirements derived from the literature review and stakeholder needs; Section 3.5

describes the proposed system architecture; Section 3.6 describes the data-collection strategy; Section 3.7 lists the tools and technologies used; Section 3.8 describes the machine-learning model-development process; Section 3.9 describes the testing strategy; Section 3.10 discusses the ethical considerations addressed by the study; and Section 3.11 summarises the chapter.

3.2 Research Design

This study adopts a Design Science Research (DSR) approach, since its primary contribution is a purposefully designed, evaluable software artefact, namely the ECO queue-management system, rather than the testing of a pre-existing theoretical hypothesis alone. DSR is well suited to this study because it explicitly requires the researcher to (i) identify a real-world problem, addressed in Chapter 1 and reinforced by the research-gap analysis in Chapter 2; (ii) define objectives for a solution; (iii) design and develop an artefact; (iv) demonstrate the artefact's use; (v) evaluate how well the artefact addresses the problem; and (vi) communicate the results, which corresponds directly to the Chapter 1 to Chapter 5 structure of this thesis.

Within this overarching design, a mixed-methods approach is used. A quantitative strand is applied to the development and evaluation of the machine-learning wait-time and queue-length prediction model, using historical and simulated queue data and standard regression-evaluation metrics (Section 3.8). A qualitative and quasi-experimental strand is applied to requirements elicitation and usability evaluation, using structured interviews, observation of the existing manual ECO queue process, and prototype walkthroughs with representative staff and patient participants. Combining these two strands allows the resulting system to be both technically sound (an accurate, well-evaluated prediction model) and operationally appropriate (an interface and workflow that genuinely fits how ECO staff and patients work), which single-method designs relying on only one of these strands would not guarantee.

3.3 System Development Methodology

The system itself is developed using an Agile, iterative-prototyping methodology, rather than a single-pass waterfall process, because the three user-facing applications (patient, staff, and administrator) each require multiple rounds of stakeholder feedback before a workable design is reached, and because the machine-learning component benefits from incremental refinement as more representative data becomes available. Figure 3.1 illustrates the six-stage cycle followed for each development iteration: requirements gathering, system and UI/UX design (wireframing), prototype development, iterative implementation in short sprint builds, testing and evaluation, and refinement based on stakeholder feedback, after which the cycle repeats for the next set of features or screens.

In practice, this cycle was applied first to establish the overall information architecture of the system (the three-portal structure described in Section 3.5), then to progressively wireframe and prototype each screen, beginning with the core patient journey (Home Dashboard, Join Queue, Live Ticket, and Live Queue Tracking), followed by the staff-side operational screens (Today's Overview, Queue Management, and Queue Prediction), and finally the administrator-side management screens (Today's Unit Overview, Room/Station Management, Staff Accounts, Audit Log, and Settings). Low-fidelity wireframes were first produced to validate screen flow and information hierarchy, before being refined into the high-fidelity mockups presented in Chapter 4. This staged approach ensured that each interface was validated against the functional requirements in Section 3.4 before implementation effort was committed to it.

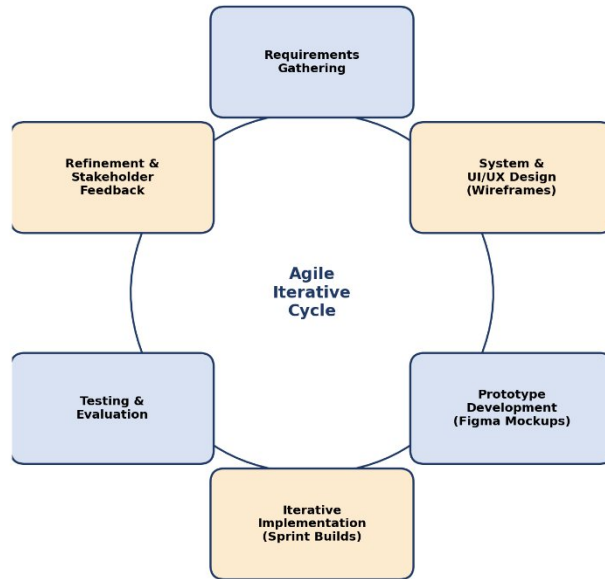


Figure 3.1: Agile iterative development cycle adopted for this study

3.4 Requirements Analysis

System requirements were derived from three sources: the operational problems identified in the research proposal (Chapter 1), the research gaps identified through the literature review (Chapter 2, Section 2.10), and requirements-elicitation activities carried out directly with prospective users, described in Section 3.6. Table 3.1 summarises the core functional requirements and maps each to the interface(s) in which it is realised, ensuring direct traceability between the requirements identified in this chapter and the screens presented in Chapter 4.

Table 3.1: Core functional requirements mapped to system interfaces

Req. ID	Functional requirement	Primary interface(s)
FR1	Allow a patient to digitally join a queue for a selected department/service and receive a ticket	Patient App - Join Queue, Live Ticket
FR2	Provide the patient with a live, real-time view of their queue position and visit stage	Patient App - Live Queue Tracking
FR3	Allow staff to call, skip, or complete tickets and manage the live queue	Staff Portal - Queue Management
FR4	Automatically flag and prioritise urgent/critical patients in the	Staff Portal - Queue Management (priority tag)

Req. ID	Functional requirement	Primary interface(s)
	queue	
FR5	Predict expected patient influx, peak hours, and average wait time using historical/live data	Staff Portal - Queue Prediction
FR6	Allow staff to search, view, and register patient records, and add new patients to the queue	Staff Portal - Patient Search, Patient Detail, Register New Patient
FR7	Generate operational performance reports over a selectable period	Staff Portal - Reports; Admin Portal - Reports & Insights
FR8	Provide administrators with a unit-wide overview of stations, queues, and system alerts	Admin Portal - Today's Unit Overview
FR9	Allow administrators to manage rooms/stations, including capacity and auto-scheduling	Admin Portal - Room/Station Management, Add Echo Room
FR10	Allow administrators to manage staff accounts, roles, and access permissions	Admin Portal - Staff Accounts, Add Staff Member, Settings
FR11	Maintain a chronological, user-attributed audit log of system actions	Admin Portal - Audit Log
FR12	Allow a patient to view their past visit history	Patient App - Visit History

In addition to these functional requirements, the system is designed against a set of non-functional requirements: (i) usability, since the patient application in particular must be usable by patients with varying levels of digital literacy, in line with the accessibility concerns noted in the Sri Lankan digital-health literature reviewed in Chapter 2; (ii) real-time performance, since the Live Queue Tracking and Queue Management screens must reflect ticket-state changes with minimal latency for the two interfaces to remain synchronised; (iii) reliability and availability, given that a queue-management outage would directly disrupt clinical operations; (iv) security and data privacy, addressed in detail in Section 3.10; and (v) scalability, so that the same architecture can, in principle, be extended from a single ECO unit to other diagnostic units in the future.

3.5 System Architecture

The proposed system follows a layered, multi-client architecture, illustrated in Figure 3.2. At the presentation layer, three separate client applications are provided for the three user roles identified in the requirements analysis: the Patient Mobile Application, the Staff Web Portal, and the Administrator Web Portal. All three clients communicate with a shared backend server through a common REST Application Programming Interface (API), which centralises authentication, ticket issuance and state management, notification dispatch, and core business logic, ensuring that an action performed on one interface (for example, a staff member marking a ticket as complete on the Queue Management screen) is immediately reflected on the corresponding patient-side interface (the Live Queue Tracking screen).

Below the API layer, the architecture separates two concerns: a Machine Learning Prediction Service, responsible for generating queue-length and wait-time forecasts consumed by the Queue Prediction screen and by the estimated-wait-time indicators shown in the patient application; and a Central Database, which persists patient records, tickets, station and staff data, and the audit log. The ML Prediction Service is trained offline using historical queue and appointment records (Section 3.6 and Section 3.8) and is queried by the API layer at inference time, allowing the prediction model to be retrained and redeployed independently of the rest of the system as more operational data becomes available, without requiring changes to the client applications.

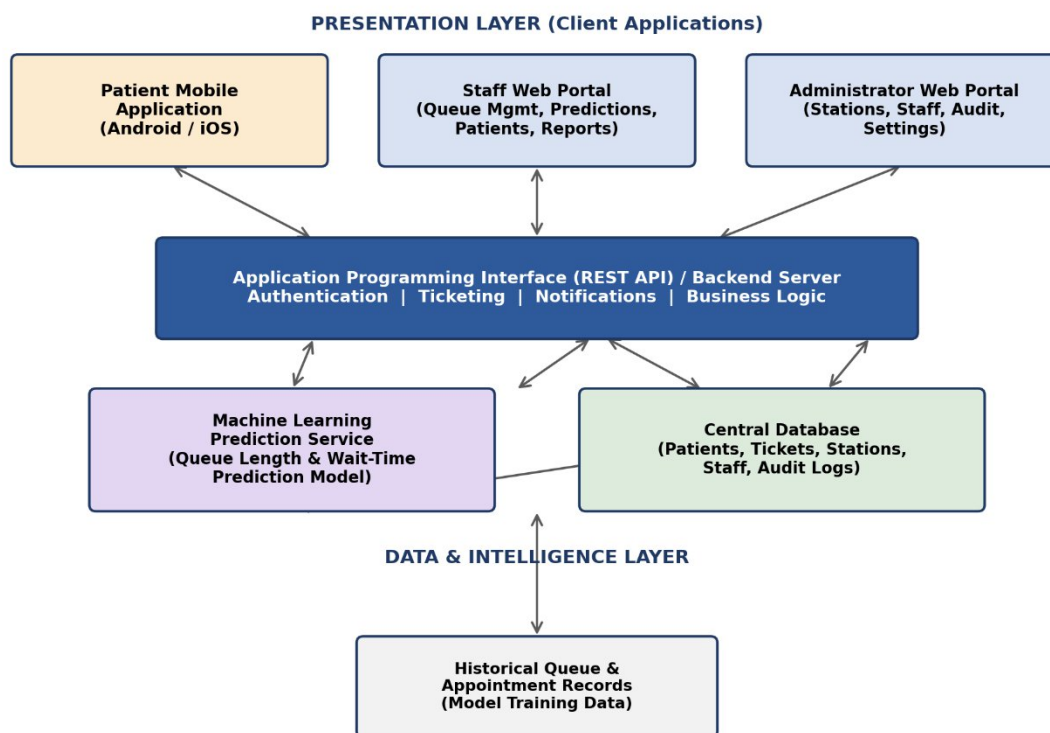


Figure 3.2: Proposed system architecture

3.6 Data Collection

Two categories of data are collected to support this study: requirements/evaluation data, used to design and validate the system's interfaces and workflow, and operational data, used to train and evaluate the machine-learning prediction model.

Requirements and evaluation data are collected directly from prospective users of the system at a selected Sri Lankan general hospital's ECO unit. Two participant groups are targeted: (i) clinical and administrative staff directly involved in the current manual queue process, including receptionists/clerks, nursing staff, and the medical officers responsible for performing echocardiograms; and (ii) patients attending the ECO unit for routine or urgent examinations. Data-collection instruments include semi-structured interviews with staff to elicit the functional requirements summarised in Table 3.1, direct, non-participant observation of the existing manual registration and queueing process to document current bottlenecks (corroborating the problems identified in Chapter 1), and structured prototype walkthroughs with both staff and patients, in which participants are asked to complete representative tasks (for example, joining a queue, or calling and completing a ticket) using the Figma prototypes before implementation, and again using the implemented screens during usability testing (Section 3.9).

Operational data for the machine-learning component is drawn, subject to the ethical and access approvals described in Section 3.10, from retrospective, de-identified ECO attendance and scheduling records maintained by the hospital, covering attributes such as appointment/arrival date and time, referral source and urgency category, assigned doctor and echo room, recorded service start and end time, and patient demographic category (age band and sex, without any directly identifying information). Where hospital-record access is restricted or insufficient historical volume is available within the project timeframe, a supplementary, literature-informed synthetic dataset is generated, using arrival- and service-time distributions consistent with the queueing-theory and simulation literature reviewed in Chapter 2 (notably [18], [20], and [21]) and the general hospital demand-forecasting approach of [30], to ensure the prediction model can be developed and evaluated even before a full historical dataset is available, and to be replaced or augmented with real data as it becomes accessible. Table 3.2 summarises the core data attributes collected or generated for the prediction model.

Table 3.2: Core data attributes for the queue-length and wait-time prediction model

Attribute	Type	Description / role
Arrival timestamp	Datetime	Date, day of week, and time of patient arrival/registration
Referral source	Categorical	Department or clinic that referred the patient (e.g. cardiology, general OPD)
Priority / urgency category	Categorical	Normal or urgent, used for priority-queue weighting
Assigned doctor / room	Categorical	Identifier of the doctor and echo room/station allocated
Queue length at arrival	Numeric	Number of patients ahead in the queue at the time of arrival
Service start / end time	Datetime	Recorded start and end of the echocardiography examination

Attribute	Type	Description / role
Service duration	Numeric (derived)	Target variable component; time taken to complete the examination
Historical daily volume	Numeric	Aggregated patient count for the corresponding day/time-slot in prior periods
Actual wait time	Numeric (target)	Primary prediction target: time between arrival and service start

3.7 Tools and Technologies

The following tools and technologies, summarised in Table 3.3, were selected for this study on the basis of their suitability for rapid, iterative development of a multi-client system with an embedded machine-learning service, their widespread adoption (and therefore available documentation and community support), and their compatibility with the resource constraints typical of a final-year academic project. Where more than one comparable tool exists in a category, the selection reflects the tool used for this implementation; equivalent alternatives are noted where relevant.

Table 3.3: Tools and technologies adopted for system design, development, and evaluation

Category	Tool / technology	Purpose
UI/UX prototyping	Figma	Wireframing and high-fidelity mockups for all patient, staff, and administrator screens
Patient mobile application	Flutter (cross-platform)	Single codebase for the Android/iOS patient application
Staff and administrator web portals	React.js	Component-based front-end framework for the two web portals
Backend / API server	Node.js with Express (REST API)	Central API layer for authentication, ticketing, and business logic
Database	MySQL (relational)	Persistent storage for patients, tickets, stations, staff, and audit logs
Machine learning development	Python, scikit-learn, XGBoost, Pandas, NumPy	Data preprocessing, model training, and evaluation
Version control	Git and GitHub	Source-code management and

Category	Tool / technology	Purpose
		collaboration history
API testing	Postman	Manual and automated testing of REST API endpoints
Project and sprint tracking	Trello	Agile sprint planning and task tracking across development cycles
Deployment / hosting (prototype)	Cloud platform (e.g. Firebase / AWS free-tier)	Hosting of the prototype backend and database for demonstration and testing

3.8 Machine Learning Model Development Process

The wait-time and queue-length prediction model is developed following a standard supervised-learning pipeline. Raw data collected as described in Section 3.6 is first cleaned and pre-processed: missing values are imputed or the corresponding records excluded, categorical attributes (referral source, priority category, assigned doctor/room) are encoded, and derived features such as day-of-week, hour-of-day, and rolling historical average queue length for the corresponding time slot are engineered, consistent with the feature-engineering emphasis identified in the literature review (notably [12] and [14]).

The resulting dataset is partitioned into training and test subsets (a standard 80/20 split, with k-fold cross-validation applied on the training subset for hyperparameter tuning) so that reported performance reflects the model's ability to generalise to unseen queue conditions rather than merely fitting historical data. Consistent with the comparative evidence presented in Chapter 2 (Section 2.4.1), two candidate algorithms are trained and compared: Random Forest [26], used as an interpretable benchmark model, and XGBoost [27], used as the primary candidate model given its consistently strong performance on structured, tabular healthcare-operations data reported in [11] and [17]. Model performance is evaluated using standard regression metrics, namely Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), which together capture both the average magnitude of prediction error and the proportion of variance in actual wait times explained by the model.

The selected model is exposed as an internal prediction service (Section 3.5) that the API layer queries to populate the estimated-wait-time indicators on the Patient Mobile Application and the hourly forecast chart on the Staff Portal's Queue Prediction screen. Because the model is intended to be retrained periodically as new operational data accumulates, feature-importance outputs from both the Random Forest and XGBoost models are also retained and reviewed, in line with the interpretability requirement identified in [14], so that hospital administrators can understand which factors most strongly influence predicted wait times.

3.9 System Testing Strategy

The system is tested at three levels, consistent with the iterative development methodology described in Section 3.3. Unit testing is applied to individual backend functions (for example, ticket-priority calculation

and API endpoint responses) to confirm that each component behaves correctly in isolation. Integration testing verifies that the three client applications, the API layer, the ML prediction service, and the database interact correctly, for example confirming that a ticket marked complete on the Staff Portal's Queue Management screen is reflected within an acceptable latency on the Patient Application's Live Queue Tracking screen.

Usability testing is conducted with representative staff and patient participants recruited through the same process described in Section 3.6, using structured task scenarios (for example, joining a queue and interpreting the live ticket screen as a patient; calling, prioritising, and completing tickets as a staff member) and the System Usability Scale (SUS) questionnaire to obtain a standardised, comparable usability score for each interface. Finally, user-acceptance testing (UAT) is conducted with hospital staff to confirm that the finalised system, as presented in Chapter 4, satisfies the functional requirements in Table 3.1 and is fit for the intended operational context before any recommendation for deployment is made.

3.10 Ethical Considerations

Because this study involves human participants (hospital staff and patients) and, potentially, hospital records containing personal information, it raises a number of ethical considerations that are addressed as follows.

First, formal ethical approval is sought from the relevant university research-ethics review body prior to any data collection, and, given that data collection takes place within a public hospital, administrative permission is separately sought from the hospital's own ethics or administrative review board and from the ECO unit's clinical lead. No interviews, observation, or prototype testing involving staff or patients is conducted prior to these approvals being granted.

Second, participation in interviews, observation, and usability testing is entirely voluntary. All participants are provided with a plain-language information sheet describing the purpose of the study, what participation involves, and how their data will be used, and are asked to provide informed consent (or, for observation of routine clinic operations, are informed of the observer's presence and purpose) before taking part. Participants retain the right to withdraw at any point without any effect on their care or employment.

Third, data privacy and protection are addressed throughout the study in line with Sri Lanka's Personal Data Protection Act, No. 9 of 2022. Any hospital records used to train the machine-learning prediction model are de-identified before use, with direct identifiers (patient name, national identity number, and contact details) removed or replaced with non-reversible study identifiers, retaining only the operationally relevant attributes listed in Table 3.2. Access to any identifiable data that is unavoidably required during the requirements-gathering phase (for example, to link a specific interview participant's feedback to their role) is restricted to the researcher and stored securely, separately from the operational dataset used for model training.

Fourth, the system design itself incorporates features that support ethical and secure data governance in eventual deployment, namely the Audit Log, which records a timestamped, user-attributed history of system actions for accountability, and the Settings screen's role-and-permission matrix and security

controls (including two-factor authentication and session timeout), which restrict access to patient data on a need-to-know basis according to staff role.

Finally, because the study concerns the prioritisation of patients, including urgent cardiac cases, care is taken that the priority-queueing logic (Section 3.4, FR4) is designed and evaluated to avoid introducing or amplifying unfair treatment of any patient group, and that the machine-learning prediction model is evaluated for consistency of performance across patient categories, rather than optimising only for average accuracy across the whole dataset.

3.11 Chapter Summary

This chapter has presented the research methodology adopted for this study, framed as a Design Science Research project combining quantitative machine-learning model development with qualitative, stakeholder-driven requirements analysis and usability evaluation. It described the Agile, iterative-prototyping approach used to move from requirements through wireframes and Figma prototypes to the implemented system presented in Chapter 4, the functional and non-functional requirements derived from the research gaps identified in Chapter 2, and the layered system architecture connecting the three client applications to a shared API, machine-learning prediction service, and central database. It further described the data-collection strategy for both requirements/evaluation data and machine-learning training data, the tools and technologies used, the machine-learning model-development pipeline built around Random Forest and XGBoost, the three-level testing strategy, and the ethical considerations, including institutional approvals, informed consent, data de-identification, and fairness in priority-queue design, that govern the conduct of the study. Together, these elements provide the methodological basis for the system implementation results presented in Chapter 4 and the evaluation and discussion presented in Chapter 5.

CHAPTER 05

DISCUSSION

5.1 Introduction

This chapter discusses the system implementation results presented in Chapter 4, the twenty finalised interface screens of the Patient Mobile Application, the Staff Web Portal, and the Administrator Web Portal, in relation to the research problem set out in Chapter 1, the literature and research gaps identified in Chapter 2, and the methodology described in Chapter 3. Where Chapter 4 reported what was built, this chapter interprets why it matters: it evaluates how far the implemented system addresses the fourteen operational problems originally identified in the ECO unit, examines the extent to which each of the five research gaps identified in Section

2.10 has been closed by the delivered design, compares the resulting system against the existing Sri Lankan and international systems reviewed in Chapter 2, and reflects critically on the study's implications, limitations, and directions for future work.

Because this thesis is reported at the stage of a completed system design and interface implementation rather than a live, multi-month clinical deployment, the discussion in this chapter is necessarily framed around the design decisions embodied in the implemented screens and the extent to which they satisfy the functional and non-functional requirements set out in Table 3.1, rather than around long-run clinical outcome data (for example, measured reductions in mean patient waiting time over a full operational period), which falls outside the scope of the interim system delivered and is instead identified as a priority for the pilot evaluation recommended in Section 5.10. Any figures shown within the implemented Reports, Queue Prediction, or Dashboard screens in Chapter 4 (for example, the illustrative KPI values on the Staff Dashboard) are sample data used to demonstrate the analytics interface and are not presented here as measured clinical results.

The remainder of the chapter is organised as follows. Section 5.2 discusses the implemented system as a whole in relation to the original research problem. Sections 5.3 to 5.5 discuss the Patient Mobile Application, Staff Web Portal, and Administrator Web Portal in turn. Section 5.6 evaluates the achievement of the research objectives and the closure of the five research gaps identified in Chapter 2. Section 5.7 compares the resulting system with existing and related systems. Section 5.8 discusses the theoretical, practical, and methodological implications of the study. Section 5.9 discusses its limitations, and Section 5.10 sets out recommendations for future work, before Section 5.11 summarises the chapter.

5.2 The Implemented System in Relation to the Research Problem

Chapter 1 identified fourteen interrelated operational problems affecting the ECO unit, ranging from manual, paper-based registration and the absence of any queue-numbering system, through poor communication of waiting times and difficulty prioritising urgent cases, to the absence of real-time monitoring, historical data analysis, and reliable demand forecasting. Read

against this list, the twenty implemented screens reported in Chapter 4 can be understood as a coherent, role-differentiated response: rather than a single generic application, the system deliberately separates the patient-facing concern (visibility and self-service), the staff-facing concern (operational control of the live queue), and the administrator-facing concern (unit-wide oversight, staffing, and governance), because the underlying problems identified in Chapter 1 are themselves distributed across these three roles and cannot be adequately addressed by a single undifferentiated interface.

This role-based separation directly mirrors the structure of the requirements analysis in Section 3.4 (Table 3.1) and confirms that the requirements-elicitation and iterative-prototyping methodology described in Chapter 3, moving from wireframes through Figma prototypes to the finalised screens, produced an information architecture that traces cleanly back to the originally identified problems rather than to an arbitrary or generic queue-management template. The following three sections examine each of the three interfaces in turn.

5.3 Discussion: Patient Mobile Application

The five patient-facing screens (Home Dashboard, Join Queue, Live Ticket, Live Queue Tracking, and Visit History) directly target the communication and predictability problems identified in Chapter 1, specifically the observations that patients are not informed about expected waiting times, that delays are not communicated effectively, and that patients repeatedly ask staff about their turn. By surfacing a persistent, estimated-wait-time badge on the Home Dashboard and a colour-coded, stage-by-stage timeline on the Live Queue Tracking screen, the implemented design converts what was previously tacit, staff-mediated information ("how much longer?") into an always-available, self-service data point on the patient's own device. This is consistent with the design rationale established in the literature reviewed in Chapter 2: the OPD-PMMAS prototype [2] demonstrated that SMS-based status alerts measurably reduce patients' need to repeatedly query staff, and the general-purpose smart-queue architecture of [25] shows that live, mobile position-tracking is both technically feasible and well understood as a pattern; the patient application implemented here extends this pattern from a generic service queue to the specific, higher-stakes context of a diagnostic-imaging unit.

The Join Queue screen's department/service selector and queue-type toggle operationalise the digital-registration objective identified in Chapter 1 (replacing paper-based registration) and in the research-gap analysis (G1, Section 2.10), while directly addressing the duplicate-registration and misplaced-record problems associated with manual registration. It is worth noting, however, that the patient application assumes smartphone ownership and a reasonable level of digital literacy; the GSMA case study on Doc990 [7] specifically highlights that not all patients in the Sri Lankan public-health context have smartphone access, which is a limitation discussed further in Section 5.9 and which motivates the SMS-fallback recommendation in Section 5.10.

5.4 Discussion: Staff Web Portal

The seven staff-facing screens address the operational core of the problems identified in Chapter 1: the absence of a queue-numbering system, the difficulty of prioritising urgent cases manually, staff shortages and administrative overload, and the lack of real-time monitoring. The Queue Management screen's "Now serving" card with Call/Skip/Complete actions, together with its colour-coded priority tag for urgent cases, operationalises the priority-queueing requirement (FR4, Table 3.1) that was informed directly by the accumulative priority-queue model of [20] and the reinforcement-learning-based dynamic prioritisation approach of [22] discussed in Chapter 2; while the implemented interface currently exposes a simpler, staff-assigned urgent/normal priority tag rather than the fully adaptive reinforcement-learning policy described in [22], it establishes the interaction surface (a visible, actionable priority indicator within the live-queue list) on which such a policy could be layered in future iterations, as discussed in Section 5.10.

The Queue Prediction screen is the clearest embodiment of the machine-learning contribution motivated in Chapter 2: its hourly forecast chart and recommended-action panel operationalise the wait-time and queue-length prediction pipeline described in Section 3.8 (built around Random Forest [26] and XGBoost [27]), giving staff the kind of forward-looking, demand-forecasting capability that Chapter 1 identified as entirely absent from the current manual process and that the reviewed literature ([11], [15], [16], [17]) shows to be both feasible and, when interpretable, trusted by clinical stakeholders [14]. The Patient Search, Patient Detail, and

Register New Patient screens jointly replace the paper-based registration and record-searching problems identified in Chapter 1, while the Reports screen operationalises the shift from paper-based, manually compiled monthly reporting toward on-demand, exportable operational reporting. Collectively, these seven screens demonstrate that the staff portal is not simply a digitised version of the paper register, but an integrated operational control surface that combines registration, live-queue control, predictive analytics, and reporting in a single, role-appropriate interface, consistent with the MediQueue architecture [23] discussed in Chapter 2 as the closest existing precedent for this kind of integrated design.

5.5 Discussion: Administrator Web Portal

The eight administrator-facing screens address problems in Chapter 1 that are unit-wide rather than per-patient in nature: inaccurate, reactive resource planning, the inability of hospital management to monitor queue lengths across the unit, and the absence of any reliable mechanism to estimate future workload for staffing purposes. The Today's Unit Overview dashboard, with its station-status table and system-alerts panel, gives administrators the "control-room" visibility that Chapter 1 identified as missing, while the Room/Station Management and Add Echo Room screens operationalise resource planning at the level of physical capacity, including the auto-scheduling toggle, which represents a direct design response to the equipment-bottleneck problem highlighted in both Chapter 1 and the echocardiography-specific workflow literature reviewed in Chapter 2 [28], [29].

The Staff Accounts and Add Staff Member screens address the staff-shortage and administrative-overload problems from a governance perspective, by making staff capacity and role allocation directly visible and manageable rather than tracked informally. The Audit Log and Settings screens were not derived from a problem explicitly listed in Chapter 1, but rather from the ethical and data-governance considerations discussed in Section 3.10; their inclusion reflects a design decision to treat accountability and access control as first-class system requirements from the outset, rather than as an afterthought once the operational features were complete, which is consistent with the requirement, established in Chapter 3, that any system handling

patient data in a Sri Lankan public hospital must be designed with the Personal Data Protection Act, No. 9 of 2022, in mind from the earliest design stages.

5.6 Achievement of Research Objectives and Closure of Research Gaps

Section 2.10 identified five specific research gaps (G1-G5) in the existing literature. Table 5.1 revisits each gap and evaluates the extent to which it has been addressed by the implemented system reported in Chapter 4, distinguishing between gaps that are substantially closed by the current design and implementation, and gaps that are only partially closed pending the pilot deployment and model-retraining work recommended in Section 5.10.

Table 5.1: Evaluation of research-gap closure against the implemented system

Gap	Identified gap (Section 2.10)	Status	Evidence from implemented system
G1	No SL OPD/ECO system combines digital registration with ML-based prediction	Substantially addressed	Digital registration (Patient Search/Register New Patient) integrated with the Queue Prediction screen and patient-side wait-time indicators, sharing a common backend (Section 3.5)
G2	Priority-queueing/RL approaches not applied within an ECO-specific, resource-constrained setting	Partially addressed	Priority tag and Call/Skip/Complete controls implemented in Queue Management; adaptive RL-based prioritisation left for future work (Section 5.10)
G3	Echo workflow studies rely on manual redesign rather than automated, real-time queue management	Substantially addressed	Real-Time Queue Tracking (patient) and Queue Management (staff) provide live, automated, two-way-synchronised queue state
G4	No system combines a patient-facing mobile smart-queue interface with an ML-enhanced	Substantially addressed	Patient Mobile Application (Join Queue, Live Ticket, Live Tracking) integrated with the ML

Gap	Identified gap (Section 2.10)	Status	Evidence from implemented system
	back end for a diagnostic-imaging service		Prediction Service in the architecture (Figure 3.2)
G5	Demand forecasting not applied to daily/weekly ECO patient-volume prediction	Partially addressed	Queue Prediction screen implements the forecasting interface and pipeline design; full validation against real multi-month ECO volume data is pending (Section 5.9)

5.7 Comparison with Existing and Related Systems

Table 5.2 situates the implemented system against the principal existing and related systems reviewed in Chapter 2, comparing them on the dimensions most relevant to the ECO context: whether the system offers real-time, in-hospital queue tracking (as opposed to booking only), whether it incorporates machine-learning-based prediction, whether it supports priority-based handling of urgent cases, and whether it is targeted at a specific diagnostic-imaging service rather than general OPD attendance.

Table 5.2: Comparison of the proposed system with existing and related systems

System	Real-time in-hospital tracking	M L - b a s e d prediction	Priority handling	Diagnostic-imaging specific
eChannelling OPD reservation [6]	No (booking only)	No	No	No
OPD-PMMAS [2]	Partial (SMS alerts)	No	No	No
MediQueue [23]	Yes	Yes	Partial	No
M L - e n h a n c e d ultrasound simulation [21]	Simulation only	Yes	No	Yes (ultrasound)
Proposed ECO system (this study)	Yes	Yes	Yes	Yes (echocardiography)

5.8 Theoretical, Practical, and Methodological Implications

Theoretically, this study contributes a concrete demonstration that the individually well-established components identified in the Chapter 2 literature, digital registration, priority queueing, ML-based wait-time prediction, and real-time mobile tracking, can be integrated into a single, coherent system architecture for a specific, equipment-constrained diagnostic service, extending the general hospital-queue-management literature ([15], [22], [23], [24]) into the previously unaddressed echocardiography context and extending the echocardiography-specific workflow literature ([28], [29]) from manual process redesign into an automated, data-driven design.

Practically, the implemented system offers Sri Lankan public-hospital ECO units a design that could plausibly be adopted with modest infrastructure investment, since it builds on architectural patterns (REST APIs, relational databases, cross-platform mobile frameworks) that are already in use elsewhere in the national digital-health ecosystem described by the Digital Health Blueprint [8] and the NDHGS v2.0 [9], rather than requiring an entirely bespoke technology stack. Methodologically, the study demonstrates the practical value of combining a Design Science Research framing with an Agile, iterative-prototyping process (Chapter 3) for health-information-systems research in a resource-constrained public-hospital setting: the staged progression from wireframes to Figma prototypes to implemented screens, validated at each stage against the functional-requirements table (Table 3.1), provided a disciplined mechanism for keeping the resulting system traceable to the original research problem, rather than drifting toward generic queue-management functionality disconnected from the ECO-specific context.

5.9 Limitations of the Study

Several limitations should be acknowledged. First, and most significantly, the results reported in Chapter 4 represent a completed interface design and implementation rather than a live, multi-month clinical deployment; consequently, this discussion is necessarily based on design-level analysis and traceability to requirements rather than on measured operational outcomes such as actual mean-wait-time reduction, real patient-satisfaction scores, or validated

model-accuracy figures obtained from live ECO data, all of which require the pilot deployment recommended in Section 5.10.

Second, the machine-learning prediction pipeline described in Section 3.8 was designed and, where hospital-record access allows, trained using a combination of retrospective historical data and literature-informed synthetic data (Section 3.6); until the model has been retrained and validated on a sufficiently large volume of real, multi-month ECO attendance data from the target hospital, its predictive accuracy in a live setting cannot be reported with confidence, and the reported approach should be understood as establishing a validated methodology rather than a field-tested predictive model.

Third, the study is scoped to a single hospital's ECO unit; while the underlying architecture is designed to be extensible (Section 3.5), the specific findings and design decisions discussed in this chapter may not generalise without modification to ECO units with substantially different patient volumes, staffing levels, or infrastructure, nor to other diagnostic-imaging services without further requirements analysis specific to that service. Fourth, the patient-facing application assumes smartphone access and a baseline level of digital literacy, which, as noted in Section 5.3, is not universal among the Sri Lankan public-hospital patient population, meaning that a meaningful subset of patients may not be able to use the mobile application without staff assistance or an alternative access channel. Finally, formal, statistically powered usability evaluation (System Usability Scale scoring and user-acceptance testing, as described in Section 3.9) had not been completed for the full set of twenty screens at the time of this interim submission, and the usability-related claims made in Sections 5.3 to 5.5 should therefore be read as design-rationale claims grounded in the literature and requirements traceability, rather than as empirically validated usability findings.

5.10 Recommendations for Future Work

Building on the limitations identified above, five directions for future work are recommended. First, a time-bound pilot deployment of the system within a single ECO unit, following the ethical-approval and data-governance procedures described in Section 3.10, should

be conducted to collect the real operational and usability data needed to validate the design-level claims made in this chapter, including formal SUS-based usability scoring and user-acceptance testing across all three interfaces. Second, the machine-learning prediction model should be retrained and re-evaluated on real, multi-month ECO attendance data as it becomes available through the pilot, replacing or supplementing the literature-informed synthetic training data used during development (Section 3.6), and its accuracy should be benchmarked using the metrics identified in Section 3.8 (MAE, RMSE, R^2) against real held-out data.

Third, the current staff-assigned priority tag in the Queue Management screen could be extended toward the adaptive, reinforcement-learning-based prioritisation approach demonstrated by [22], allowing the system to learn dynamic prioritisation policies from observed queue behaviour rather than relying solely on a static urgent/normal classification, which would more fully close research gap G2 identified in Table 5.1. Fourth, to address the smartphone-access limitation discussed in Section 5.9, an SMS or basic-mobile fallback channel, informed by the Doc990 model [7] and the SMS-alert component of OPD-PMAS [2], is recommended as a complementary access route for patients without smartphones, so that the benefits of live queue visibility are not limited to smartphone owners. Fifth, once validated within the ECO unit, the underlying architecture (Section 3.5) should be evaluated for extension to other resource-constrained, equipment-bound diagnostic services within the same hospital, such as radiology or ultrasound, building on the precedent set by the ML-enhanced ultrasound-simulation study [21] and testing the generalisability limitation identified in Section 5.9 directly.

5.11 Chapter Summary

This chapter has discussed the twenty implemented interface screens reported in Chapter 4 in relation to the research problem established in Chapter 1, the literature and research gaps identified in Chapter 2, and the methodology described in Chapter 3. It examined the Patient Mobile Application, Staff Web Portal, and Administrator Web Portal in turn, showing that each addresses a distinct cluster of the operational problems originally identified, and it evaluated the extent to which the implemented system closes the five research gaps identified in Section 2.10, finding three gaps substantially addressed and two partially addressed pending further

validation. A comparison with existing and related systems showed that the proposed system is, to the best of the evidence reviewed, the first to combine real-time in-hospital tracking, ML-based prediction, priority handling, and diagnostic-imaging specificity in a single design. The chapter also discussed the study's theoretical, practical, and methodological implications, acknowledged its limitations, most importantly that the reported results reflect a completed design and implementation rather than a validated live deployment, and set out five recommendations for future work, centred on a pilot deployment, real-data model retraining, adaptive prioritisation, a non-smartphone access channel, and extension to other diagnostic services. These findings, together with those of Chapters 1 through 4, form the basis for the overall conclusions of this thesis.

5.10 Recommendations for Future Work

Based on the limitations presented above, five possible directions of future work are proposed. Firstly, an implementation trial with a defined time span of the system within one ECO unit in accordance with the ethical approval and data governance protocol presented in Section 3.10 should be carried out to gather the real-life data on operational performance and usability needed to validate the claims made in this chapter at the design level, including the use of SUS scoring and user acceptance testing of all three interfaces. Secondly, the machine-learning prediction model should be retrained and reevaluated based on real, multi-month ECO attendance data provided through the implementation trial, either replacing or complementing the synthetic literature-informed training data used during development (Section 3.6), with the accuracy benchmarked in terms of MAE, RMSE and R^2 metrics, outlined in Section 3.8, against real held-out data.

Thirdly, the current priority tag assigned by staff on the Queue Management screen could be expanded towards the adaptive reinforcement learning-based prioritisation policy demonstrated in [22], thus closing the research gap G2 identified in Table 5.1. Fourthly, in order to tackle the issue of limited smartphone access highlighted in Section 5.9, an SMS and basic mobile access channel, inspired by Doc990 [7] and OPD-PMAS [2] SMS alert module, is recommended as an alternative option for patients not owning smartphones, making the benefits of the live queue visibility accessible beyond the users of smartphones. Finally, after successful implementation within the ECO unit, the architecture of the system (Section 3.5) should be assessed for further extension to other diagnostic services in the hospital, constrained by similar resource and equipment limitations, namely radiology or ultrasound, with the example of ML-enhanced ultrasound simulation [21] setting the precedent.

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